The Relationship Between University Students' Sleep and Caffeine Intake - A Correlational Study

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PSY 290: Research Methods

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April 17, 2024

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The National Sleep Foundation recommends 8 to 10 hours of sleep for teenagers, 7 to 9 hours of sleep for young adults and adults, and 7 to 8 hours of sleep for older adults to be healthy and not suffer from sleep disorders (Hirshkowitz et al., 2015). However, according to a study done by C.Y. Song and W.S. Wong (2010) among college students in Hong Kong, 68.6% of the students were insomniacs. Poor sleep among college students can result in increased tension, irritation, depression, confusion, lower life satisfaction, or poor academic performance (Yves, 2022).

One of the reasons students are not getting enough sleep is due to high caffeine intake. Many societal groups, including children, adolescents, and adults, often drink caffeinated beverages. Coffee is the primary source of caffeine consumption for adults in the United States, where the average daily intake is 4 mg per kg body weight (Melanie et al., 2010). Caffeine is a naturally occurring stimulant of the central nervous system that belongs to the methylxanthine class. It is mostly derived from coffee beans, although it can also be found in some teas and cacao beans. It is also an additive to soda and energy drinks (Justin et al., 2023).

The human body responds to caffeine in both positive and negative ways, and it affects many different systems, such as the immunological system, digestive tract, respiratory system, central nervous system, and urinary tract. According to Rodak and colleagues (2021), these effects vary depending on the product type and quantity of caffeine as well as individual characteristics (e.g., age, diet, and sex). Caffeine is the most widely used stimulant to counteract the effects of sleepiness, but it also has detrimental effects on subsequent sleep (Francesca et al., 2013). According to a study conducted by Rebecca and colleagues (2006), adolescents who use large amounts of caffeine are more likely to feel exhausted in the morning and have problems sleeping. Furthermore, numerous studies with large sample sizes and population-based research show a link between regular daily dietary caffeine intake and sleep disturbances as well as daytime sleepiness (Timothy, 2008).

The Current Study

The primary aim of this study is to understand the relationship between caffeine intake and sleep among university students. Given previous studies (Rebecca 2006; James, 1998), we hypothesize that students with higher caffeine intake are more likely to sleep less per day. Data was collected through the distribution of surveys to PSY 290 students and other university students. The data we gathered consists of age, gender, race/ethnicity of the students, the amount of water, caffeine they consume, whether they exercise, and the amount of sleep they get per day on average. We conducted an ANOVA analysis to examine if the mean amount of sleep differs by the different amounts of caffeine intake.

Methods

Participants

We aimed to collect a total of at least 80 participants and expected to end with the final sample of 70 participants who met the inclusion criteria and completed the survey. This number was obtained by conducting a power analysis using an expected large effect size of 0.4 (James, 1998; Rebecca et al., 2006), desired power of 80%, and alpha value of .05.

In order to participate in this experiment, which was conducted in March 2024, the participants had to be 18 or older. PSY 290 students from two classes were recruited to complete the study via a course announcement and they were encouraged to forward the study's link to any other university students they knew. They received no compensation for their participation in the study. This study did not require an IRB review because it was part of a course project. We followed all relevant ethical rules, including providing subjects with consent forms.

Measures

The survey's first three questions had questions about demographics. Participants provided information on their age, gender, race/ethnicity. The question about age was a free response. For gender, participants had the option of selecting male, female, non-binary, and prefer not to say. Race/ethnicity categories were White/Caucasian, Native American, Hispanic/ Latinx, Black/African American, and Asian. Next was a question asking about their caffeine intake, and they were given three choices: less than one per day, one per day, and more than one per day. The last three questions were about their water consumption in liters (free response), whether they exercise, and how many hours they sleep daily on average (free response).

Procedures

In this experiment, we distributed an online survey, and students received the link to the survey via their student email and other social media applications like WhatsApp, and Instagram. Students were asked to complete the study at a time and location of their choosing using the electronic devices available to them. Students read a detailed description of the study and signed a consent form. The study consisted of eight questions and was estimated to take three to five minutes on average.

This is a correlational study. The participants answered the first three questions about their age, gender, and race/ethnicity. Then they were asked questions about their caffeine intake, sleep, exercise, and water consumption. After collecting the data, we checked for outliers, data points further than 3sd from the mean, and the shape of the distribution of sleep. All of our questions were compulsory, so everyone answered all of them. We restricted the range for sleep (0 -24 hours).

Results

Sample Descriptives

We recruited 52 participants between March 28 and April 9. Of these participants, 1 was removed due to the age of the participant (below 18), leading to a total sample size of 51 participants. The mean age of this study was 19.765 (*SD* = 2.337), with 35.294% of participants self-identifying as White/Caucasian (N = 18), 0% as Native American (N = 0), 11.765% as Hispanic/ Latinx (N = 6), 3.922% as Black/African American (N = 2), 45.098% as Asian (N = 23), and 3.922% as other (N = 2). In terms of gender breakdown, 54.902% of participants were female (N = 28), 41.176% were male (N = 21) and 3.922% were non-binary students (N = 2).

Among participants, 62.745% reported drinking less than one caffeinated beverage per day (N = 32), 31.373% drank one caffeinated beverage per day (N = 16) and 5.882% drank two or more caffeinated beverages per day (N = 3). The average reported hours of sleep per day was 7.206 (*SD* = 1.167); the average hours of sleep per day for participants who drink less than one caffeinated beverage per day was 7.328 (*SD* = 1.060), for participants who drink one caffeinated beverage per day was 7.250 (*SD* = 1.238), and for participants who drink two or more caffeinated beverages per day was 5.667 (*SD* = 1.155). Hours of sleep per day was right skewed distributed, and the standard deviation of hours of sleep per day varied between groups. Descriptive statistics of hours of sleep per day and the number of caffeinated beverages per day are provided in Table 1.

Table 1

Descriptive Statistics

	Hours of sleep	
	Mean	SD
Less than one caffeinated beverage per day (N = 32)	7.328	1.060
One caffeinated beverage per day (N = 16)	7.250	1.238
Two or more caffeinated beverages per day (N = 3)	5.667	1.155
Total (N = 51)	7.206	1.167

Analysis of Variance (ANOVA)

An analysis of variance (ANOVA) was conducted to compare the number of hours of sleep participants get per day depending on the number of caffeinated beverages they drink per day. All analyses were done using JASP. We hypothesized that students with higher caffeine intake (i.e., more caffeinated drinks) are more likely to sleep less per day.

A one-way ANOVA showed that mean hours of sleep did not differ significantly by the number of caffeinated beverages participants drink per day [$F(2, 48) = 3.023$, $p = 0.058$].

Discussion

The purpose of this study was to look for the relationship between levels of caffeine intake and hours of sleep per day. We hypothesized that students with higher caffeine intake are more likely to sleep less per day. The hypothesis is not supported by the data collected from the study, as there is no significant difference between the hours of sleep students get per day based on the number of caffeinated beverages, they intake per day.

These findings are consistent with some previous research findings, which have shown that sleep duration was not impacted by a daily habit of consuming up to seven cups or 600 mg of caffeine per day (Sanchez et al., 2005) and that the effects of caffeine depend on the amount of caffeine consumed, the type of product it is contained in and also on the differences in individuals such as age, sex, diet, etc (Rodak et al., 2021). There are also other findings that have shown the opposite, regular daily consumption of caffeine is linked to sleep disturbances and consequent daytime drowsiness (Timothy, 2008).

Strengths and Limitations

The present study had many strengths. The sample was composed of individuals from a variety of racial and ethnic origins, so it was less prejudiced toward any one group of people and gave us a more comprehensive understanding of the world. However, limitations should be taken into account while interpreting the findings. First, our goal was to recruit at least 80 people, and we anticipated that number to reach 70. However, we were only able to recruit 51 participants, indicating that we lacked sufficient power (i.e., not enough participants). Because of the limited size of our sample, we cannot be certain of the results we obtained.

Second, the research uses a sample of university students. This is a convenience sample as it just represents university students, so there might be problems with external validity (i.e., it cannot be generalized to non-university students like the working population). This is also a snowball sample because students who are taking PSY 290 course were encouraged to share the study with their friends whom they think are suitable for this study (i.e., university students above age 18). This type of sampling might lead to inaccurate results as we do not know whether the survey was shared with participants who fit the criteria of being a university student above age 18. Third, this is an online study where the link is shared on public sites, and there is no restriction on how many times a participant can fill out the survey. This might lead to getting data from the same participant twice or thrice which leads to inaccurate results. Fourth, there is a threat to statistical validity as the dependent variable is skewed to the right.

Future Directions

The results of this study suggest that there is no relation between the amount of caffeine consumed and the number of hours of sleep a day; nevertheless, the sample size and sampling technique may have affected the findings. So, future researchers might want to design a study where the sample size is large enough (with enough power) to be sure of the results. One other potential adjustment is the sampling technique used (i.e., how the sample is collected). To improve generalizability and make the study more accurate, they can try to get an equal number of sets of people who drink less than one caffeinated beverage per day, one per day and two or per day and randomly select an equal number of participants from each set, in addition to gathering data from all types of populations instead of just using university students.

Future researchers can also change their area of focus, such as observing the quality of sleep instead of the quantity of sleep (i.e., checking whether the sleep participants get is without disturbances such as frequent nightmares, restlessness, etc), and designing experiments to find the maximum number of caffeinated beverages a student can consume without facing severe consequences. As the study found no association between the amount of caffeine consumed and the number of hours slept per day, students should not worry about the number of caffeinated beverages they intake, and universities can provide coffee at various locations across campus.

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